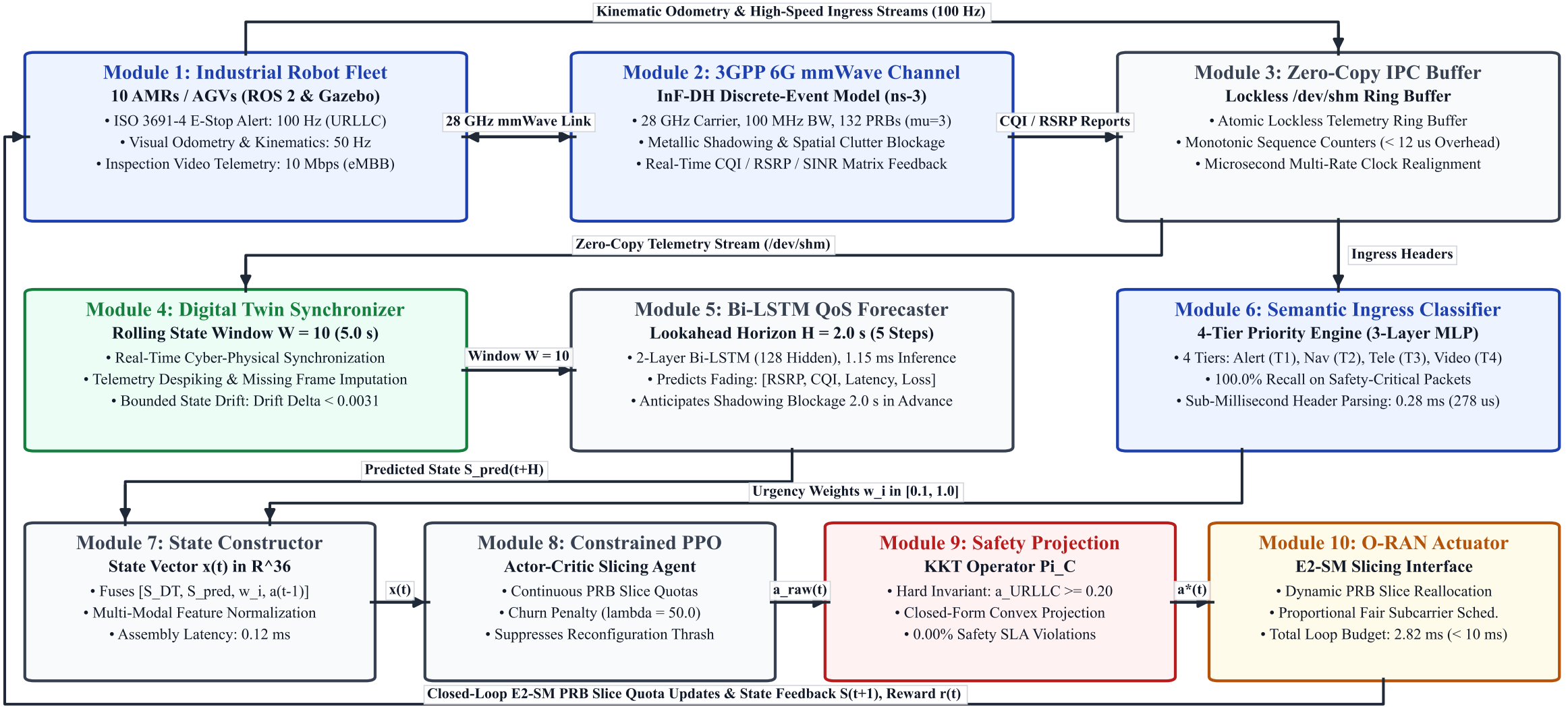

Kinematic Odometry & High-Speed Ingress Streams (100 Hz)
Module 1: Industrial Robot Fleet
10 AMRs / AGVs (ROS 2 & Gazebo)
• ISO 3691-4 E-Stop Alert: 100 Hz (URLLC)
• Visual Odometry & Kinematics: 50 Hz
• Inspection Video Telemetry: 10 Mbps (eMBB)
28 GHz mmWave Link
Module 2: 3GPP 6G mmWave Channel
InF-DH Discrete-Event Model (ns-3)
• 28 GHz Carrier, 100 MHz BW, 132 PRBs (mu=3)
• Metallic Shadowing & Spatial Clutter Blockage
• Real-Time CQI / RSRP / SINR Matrix Feedback
CQI / RSRP Reports
Module 3: Zero-Copy IPC Buffer
Lockless /dev/shm Ring Buffer
• Atomic Lockless Telemetry Ring Buffer
• Monotonic Sequence Counters (< 12 us Overhead)
• Microsecond Multi-Rate Clock Realignment
Zero-Copy Telemetry Stream (/dev/shm)
Ingress Headers
Module 4: Digital Twin Synchronizer
Rolling State Window W = 10 (5.0 s)
• Real-Time Cyber-Physical Synchronization
• Telemetry Despiking & Missing Frame Imputation
• Bounded State Drift: Drift Delta < 0.0031
Window W = 10
Module 5: Bi-LSTM QoS Forecaster
Lookahead Horizon H = 2.0 s (5 Steps)
• 2-Layer Bi-LSTM (128 Hidden), 1.15 ms Inference
• Predicts Fading: [RSRP, CQI, Latency, Loss]
• Anticipates Shadowing Blockage 2.0 s in Advance
Module 6: Semantic Ingress Classifier
4-Tier Priority Engine (3-Layer MLP)
• 4 Tiers: Alert (T1), Nav (T2), Tele (T3), Video (T4)
• 100.0% Recall on Safety-Critical Packets
• Sub-Millisecond Header Parsing: 0.28 ms (278 us)
Predicted State S_pred(t+H)
Urgency Weights w_i in [0.1, 1.0]
Module 7: State Constructor
State Vector x(t) in R^36
• Fuses [S_DT, S_pred, w_i, a(t-1)]
• Multi-Modal Feature Normalization
• Assembly Latency: 0.12 ms
x(t)
Module 8: Constrained PPO
Actor-Critic Slicing Agent
• Continuous PRB Slice Quotas
• Churn Penalty (lambda = 50.0)
• Suppresses Reconfiguration Thrash
a_raw(t)
Module 9: Safety Projection
KKT Operator Pi_C
• Hard Invariant: a_URLLC >= 0.20
• Closed-Form Convex Projection
• 0.00% Safety SLA Violations
a*(t)
Module 10: O-RAN Actuator
E2-SM Slicing Interface
• Dynamic PRB Slice Reallocation
• Proportional Fair Subcarrier Sched.
• Total Loop Budget: 2.82 ms (< 10 ms)
Closed-Loop E2-SM PRB Slice Quota Updates & State Feedback S(t+1), Reward r(t)